

Inequalities in Two Variables

AQA · KS4 Year 9–10

Answer each question in your workbook. Draw graphs carefully on squared paper.

VISUAL EXAMPLE



$x < 5$ on a number line: open circle at 5, shade everything to the left.

COMMON MISTAKES

- Solid line for $<$ or $>$ (should be dashed).
- Dashed line for $\leq$ or $\geq$ (should be solid).
- Shading the wrong side — always test a point.
- Region must satisfy ALL the inequalities.

METHOD

- 1 Draw the boundary line (sign $=$)
- 2 Dashed for $</>$, solid for $\leq / \geq$
- 3 Test a point not on the line
- 4 Shade the side that works
- 5 Several inequalities — shade the overlap

WORKED EXAMPLE

Show $y < x + 2$.
Draw $y = x + 2$ SOLID (). Test (0,0): $0 < 2$ true — shade the side with (0,0).
That shaded area below the line is the region.

VISUAL AID

On x–y axes, each inequality gives a half-plane: a boundary line (dashed for strict, solid for inclusive) divides the grid in two, and you shade the side where the inequality holds. The region for several inequalities is where all the shadings overlap.

1. (2 marks)

Solid or dashed line? a) $y > 2x$ b) $y \leq x + 1$

2. (2 marks)

Satisfies the inequality? a) (1,4) in $y > x + 2$? b) (2,1) in $y \leq 2x - 1$?

3. (3 marks)

Draw the boundary and shade the region: a) $x > 2$ b) $y \leq 3$ c) $y < x$

4. (3 marks)

Does (0,0) lie in each region? a) $y < x + 1$ b) $y \leq 2x - 3$ c) $y > x$

5. (4 marks)

Shade R satisfying $x \geq 0$, $y \geq 0$, $x + y \leq 4$. State the three corners.

6. (3 marks)

A region is above $y = 1$, right of $x = 1$, below $y = 5 - x$. Write the three inequalities.

7. (4 marks)

R: $y \leq 1$, $y \leq 2x - 1$, $x \leq 4$. a) Which lines are solid? b) Is (3, 2) inside R? Show working.

8. (5 marks)

Van: x small + y large parcels, $x + y \leq 10$, $x \geq 2$, $y \geq 3$. a) Meaning of each. b) A valid (x,y). c) Greatest total and an (x,y) achieving it.